EXERCISE SHEET
More Descriptive Statistics and Poisson distribution

EXERCISE 1 *

1000 persons express their level of satisfaction with their recently bought new car using a scale from 1 to 5 (1=very dissatisfied, 5= very satisfied). The answers are represented in the table:

Satisfaction Level	1	2	3	4	5
Number of Individuals	25	125	200	475	175

For each of the following statements decided whether it is TRUE or FALSE.

You do not need R Studio or a calculator for this.

(a) The median is 3. [TRUE] [**FALSE**]

(b) The sample distribution of Satisfaction Level is right skewed. [TRUE] [**FALSE**]

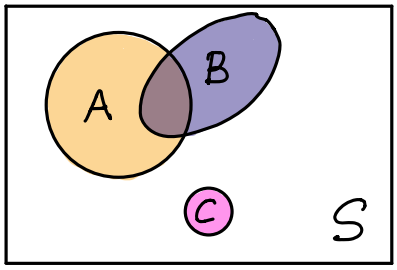
(c) The first and third quartiles are respectively 3 and 4. [**TRUE**] [FALSE]

(d) The mean is greater than 3. [**TRUE**] [FALSE]

(e) The standard deviation is greater than 2. [TRUE] [**FALSE**]

EXERCISE 2 *

Let A, B and C be 3 events as represented in the Venn diagram below.



For each of the following statements decided whether it is TRUE or FALSE.

(a) $(A \cup B)^c = C$ [TRUE] [**FALSE**]

(b) $(A \cap B)^c \supseteq C$ [**TRUE**] [FALSE]

(c) $(C^c \cap B)^c \cap A = \varnothing$ [TRUE] [**FALSE**]

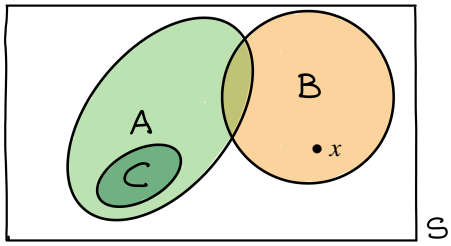
(d) $(A \cap B^c)^c = A^c \cup B$ [**TRUE**] [FALSE]

(e) $C \cap (A \cap B^c) = \varnothing$ [**TRUE**] [FALSE]

(f) $C^c \subseteq A \cup B$ [TRUE] [**FALSE**]

EXERCISE 3 *

Let A, B and C be 3 events and x an element of B as represented in the Venn diagram below.



For each of the following statements decided whether it is TRUE or FALSE.

(a) $x \in (A \backslash C)^c$ [**TRUE**] [FALSE]

(b) $(C^c \cap B) = B$ [**TRUE**] [FALSE]

(c) $x \notin (A^c \cup C)^c$ [**TRUE**] [FALSE]

(d) $A \backslash C = A \cap C^c$ [**TRUE**] [FALSE]

(e) $(A \cap B)^c = \varnothing$ [TRUE] [**FALSE**]

(f) $C^c \cap (A \cap B^c) \supseteq C$ [TRUE] [**FALSE**]

EXERCISE 4 */**

For the following questions, refer to the dataset: **UNData.RData**.

The dataset contains national Statistics from the United Nations, Mostly From 2009–2011 including national health, welfare, and education statistics data.

It includes 193 observations (countries) and 7 variables described below.

Region: Region of the world: Africa, Asia, Caribbean, Europe, Latin Amer, North America, NorthAtlantic, Oceania.

Group: A factor with levels oecd for countries that are members of the OECD, the Organization for Economic Co-operation and Development, as of May 2012, africa for countries on the African continent, and other for all other countries. No OECD countries are located in Africa.

Fertility: Total fertility rate, number of children per woman.

GDP: Per capita gross domestic product in US dollars.

LifeExp: Female life expectancy, years.

Urban: percentage of urban population with levels high, medium, low.

InfantMortaility: Infant deaths by age 1 year per 1000 live births

(a) Consider a histogram for the variable GDP and classes: [0,2000], (2000,10000], (10000,20000], (20000,150000]. For which class the frequency density is higher? Report its value. **[[0,2000], frequency density = 0.0001580311]**

(b) Consider a histogram for the variable GDP and classes: [0,2000], (2000,10000], (10000,20000], (20000,150000]. What is the absolute frequency of the class which contains the highest number of countries? **[67]**

(c) Which is the most represented region in the dataset? **[Africa, 52 observations]**

(d) What is the percentage of countries with a low percentage of urban population in the dataset? **[49.22%]**

(e) What is the percentage of countries in the dataset that are in Africa and have a low percentage of urban population? **[joint relative frequency: 19.69%]**

(f) What is the percentage of African countries in the dataset that have a low percentage of urban population? **[conditional relative frequency: 73.08%]**

(g) What is the percentage of countries with low percentage of urban population that are in Africa? **[conditional relative frequency: 40%]**

(h) In which countries life expectancy is lower than 50 years? **[Afghanistan, Lesotho, Sierra Leone, Swaziland]**

(i) Filter the dataset according to the following criteria:

- Countries that belong to the Group: "other"
- Countries that belong to the Group "other" and are in Europe
- Countries that belong to the Group "other", are in Europe and **either** have a GDP per capita greater than 25000 **or** have a high percentage of urban population **[Belarus, Iceland, Malta, Spain]**
- Set the value of the variable Group for Malta and Spain to: "oecd".

(j) Find the IQR of the variable Fertility. Find the value of Fertility over which an observation is considered an upper outlier with respect to this variable. What are the outliers with respect to the variable Fertility? **[IQR=1.95,Upper Limit=6.62, Niger]**

(k) Consider the conditional distribution of the variable Fertility given the variable Region. For which region the conditional distribution of Fertility presents the highest number of outliers? (Hint: use boxplot()) **[Asia]**

(l) Consider the median GDP per capita for each of the subgroups of countries defined by the variable Region. What is the median GDP per capita of the region with the second highest median GDP per capita? (Hint: use tapply()) **[Europe, 19599.20]**

EXERCISE 5 */**/***

After receiving a large shipment of computer chips, a sample of 800 chips is randomly selected. If 3 or fewer defective chips are found, the entire lot is accepted without inspecting the remaining chips in the lot. If 4 or more chips are defective, every chip in the entire lot is carefully inspected at the supplier's expense. Assume that the true proportion of nonconforming computer chips being supplied is 0.001, independently from each other. Use the Poisson distribution to approximate the number of faulty chips.

(a) How many faulty items do you expect on average in a shipment? [**0.8**]

(b) What is the probability of no defective chip in the sample? **[0.449329]**

(c) Calculate the probability in (a) using a Bernoulli model instead. [**0.4491491**]

(d) (Back to Poisson model) What is the probability of finding exactly 4 defective chips? [**0.007668548**]

(e) What is the probability that the entire lot will be inspected? [**0.009079858**]

(f) The supplier claims that the probability to have more than 2 defective chips in the randomly selected sample is below 5%. Is this true? [**Yes. It's 0.0474226**]

(g) Plot the probability mass function of the number of faulty items.

(h) ***Using rbinom(), simulate 100,000 times and independently the number of faulty chips found in samples of 800 chips. Store the 100,000 simulations in a vector of length 100,000. Produce the sample relative frequencies and compare them with the probability mass function of the Poisson. On the basis of this comparison, do you think the poisson distribution is a good model for this scenario? [**check video solution script**]

(i) ***Using rpoiss(), simulate 100 times and independently the number of faulty chips found in samples of 800 chips. Store the 100 simulations in a vector of length 100. Can you approximate the probability in (e) using the simulated data?
[YES, but it might not be accurate -> rare event]
Repeat the experiment using 1,000,000 simulations. How long does it take? Can you approximate the probability in (e) using the simulated data?
[**Under a second, the result should be close to 0.009. Check video solution script**]

(j) ***Assume that the company pays a fine according to the number of faulty chips found in the sample of 800 chips. In particular:

- If the faulty chips are 3 or fewer, a fine of 100 euros per faulty chip is applied.
- If the faulty chips are 4 or 5, a fine of 150 euros per faulty chip is applied.
- If the faulty chips are more than 5, the fine applied is a flat 2000 euros

Approximate through simulation the average fine paid by the company for an inspected batch. (Hint! Use the samples simulated at the previous point!).

[the result should be around 82, but the precision depends on simulations used. Note that it is possible to solve this analytically, too.]